

will weigh 0.02 pounds more for each extra calorie you consume daily, or about 2 pounds for each extra 100 calories per day. The units of $\partial w/\partial n$ are pounds per minute. The statement

$$\frac{\partial w}{\partial n}(2000, 15) = -0.025$$

means that for the same calorie consumption and number of minutes of exercise, you will weigh 0.025 pounds less for each extra minute you exercise daily, or about 1 pound less for each extra 40 minutes per day. So if you eat an extra 100 calories each day and exercise about 80 minutes more each day, your weight should remain roughly steady.

Exercises and Problems for Section 14.1

Exercises

1. Given the following table of values for $z = f(x, y)$, estimate $f_x(3, 2)$ and $f_y(3, 2)$. Assume that f is differentiable.

$x \setminus y$	0	2	5
1	1	2	4
3	-1	1	2
6	-3	0	0

2. Using difference quotients, estimate $f_x(3, 2)$ and $f_y(3, 2)$ for the function given by

$$f(x, y) = \frac{x^2}{y+1}.$$

[Recall: A difference quotient is an expression of the form $(f(a+h, b) - f(a, b))/h$.]

3. Use difference quotients with $\Delta x = 0.1$ and $\Delta y = 0.1$ to estimate $f_x(1, 3)$ and $f_y(1, 3)$ where

$$f(x, y) = e^{-x} \sin y.$$

Then give better estimates by using $\Delta x = 0.01$ and $\Delta y = 0.01$.

4. The price P in dollars to purchase a used car is a function of its original cost, C , in dollars, and its age, A , in years.
- What are the units of $\partial P/\partial A$?
 - What is the sign of $\partial P/\partial A$ and why?
 - What are the units of $\partial P/\partial C$?
 - What is the sign of $\partial P/\partial C$ and why?
5. Your monthly car payment in dollars is $P = f(P_0, t, r)$, where P_0 is the amount you borrowed, t is the number of months it takes to pay off the loan, and $r\%$ is the interest rate. What are the units, the financial meaning, and the signs of $\partial P/\partial t$ and $\partial P/\partial r$?
6. A drug is injected into a patient's blood vessel. The function $c = f(x, t)$ represents the concentration of the drug at a distance x mm in the direction of the blood flow

measured from the point of injection and at time t seconds since the injection. What are the units of the following partial derivatives? What are their practical interpretations? What do you expect their signs to be?

- (a) $\partial c/\partial x$ (b) $\partial c/\partial t$

7. You borrow $\$A$ at an interest rate of $r\%$ (per month) and pay it off over t months by making monthly payments of $P = g(A, r, t)$ dollars. In financial terms, what do the following statements tell you?

(a) $g(8000, 1, 24) = 376.59$

(b) $\left. \frac{\partial g}{\partial A} \right|_{(8000, 1, 24)} = 0.047$

(c) $\left. \frac{\partial g}{\partial r} \right|_{(8000, 1, 24)} = 44.83$

8. The sales of a product, $S = f(p, a)$, are a function of the price, p , of the product (in dollars per unit) and the amount, a , spent on advertising (in thousands of dollars).

- Do you expect f_p to be positive or negative? Why?
- Explain the meaning of the statement $f_a(8, 12) = 150$ in terms of sales.

9. The quantity, Q , of beef purchased at a store, in kilograms per week, is a function of the price of beef, b , and the price of chicken, c , both in dollars per kilogram.

- Do you expect $\partial Q/\partial b$ to be positive or negative? Explain.
- Do you expect $\partial Q/\partial c$ to be positive or negative? Explain.
- Interpret the statement $\partial Q/\partial b = -213$ in terms of quantity of beef purchased.

In Exercises 10–13, determine the sign of f_x and f_y at the point using the contour diagram of f in Figure 14.7.

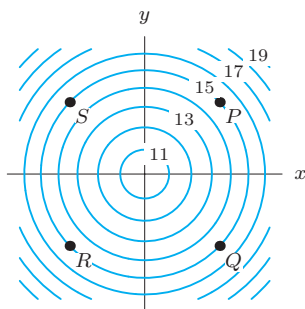


Figure 14.7

10. P 11. Q 12. R 13. S

Problems

17. Figure 14.8 is a contour diagram for $z = f(x, y)$. Is f_x positive or negative? Is f_y positive or negative? Estimate $f(2, 1)$, $f_x(2, 1)$, and $f_y(2, 1)$.

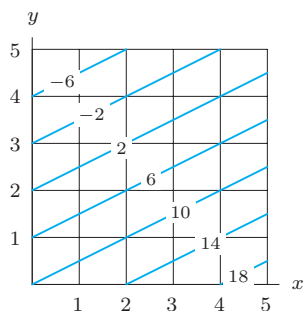


Figure 14.8

18. Approximate $f_x(3, 5)$ using the contour diagram of $f(x, y)$ in Figure 14.9.

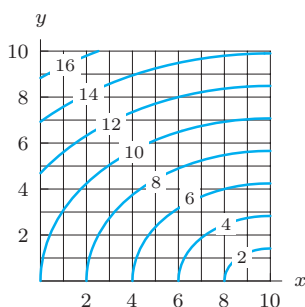


Figure 14.9

For Exercises 14–16, refer to Table 12.2 on page 672 giving the temperature adjusted for wind chill, C , in $^{\circ}\text{F}$, as a function $f(w, T)$ of the wind speed, w , in mph, and the temperature, T , in $^{\circ}\text{F}$. The temperature adjusted for wind chill tells you how cold it feels, as a result of the combination of wind and temperature.

14. Estimate $f_w(10, 25)$. What does your answer mean in practical terms?
15. Estimate $f_T(5, 20)$. What does your answer mean in practical terms?
16. From Table 12.2 you can see that when the temperature is 20°F , the temperature adjusted for wind-chill drops by an average of about 0.8°F with every 1 mph increase in wind speed from 5 mph to 10 mph. Which partial derivative is this telling you about?

19. The quantity Q (in pounds) of beef that a certain community buys during a week is a function $Q = f(b, c)$ of the prices of beef, b , and chicken, c , during the week. Do you expect $\partial Q / \partial b$ to be positive or negative? What about $\partial Q / \partial c$?

20. The average price of large cars getting low gas mileage (“gas guzzlers”) is x and the average price of a gallon of gasoline is y . The number, q_1 , of gas guzzlers bought in a year, depends on both x and y , so $q_1 = f(x, y)$. Similarly, if q_2 is the number of gallons of gas bought to fill gas guzzlers in a year, then $q_2 = g(x, y)$.

- (a) What do you expect the signs of $\partial q_1 / \partial x$ and $\partial q_2 / \partial y$ to be? Explain.
- (b) What do you expect the signs of $\partial q_1 / \partial y$ and $\partial q_2 / \partial x$ to be? Explain.

21. An experiment to measure the toxicity of formaldehyde yielded the data in Table 14.3. The values show the percent, $P = f(t, c)$, of rats surviving an exposure to formaldehyde at a concentration of c (in parts per million, ppm) after t months. Estimate $f_t(18, 6)$ and $f_c(18, 6)$. Interpret your answers in terms of formaldehyde toxicity.

Table 14.3

		Time t (months)					
		14	16	18	20	22	24
Conc. c (ppm)	0	100	100	100	99	97	95
	2	100	99	98	97	95	92
	6	96	95	93	90	86	80
	15	96	93	82	70	58	36

22. Figure 14.10 shows contours of $f(x, y)$ with values of f on the contours omitted. If $f_x(P) > 0$, find the sign of

(a) $f_y(P)$ (b) $f_y(Q)$ (c) $f_x(Q)$

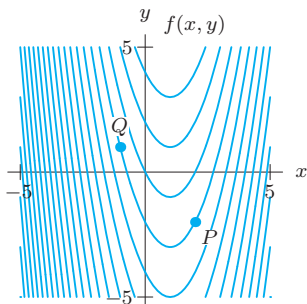


Figure 14.10

23. Figure 14.11 shows the contour diagram of $g(x, y)$. Mark the points on the contours where

(a) $g_x = 0$ (b) $g_y = 0$

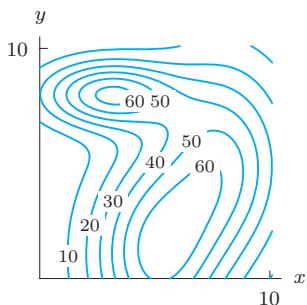


Figure 14.11

24. The surface $z = f(x, y)$ is shown in Figure 14.12. The points A and B are in the xy -plane.

- (a) What is the sign of
 (i) $f_x(A)$? (ii) $f_y(A)$?
 (b) The point P in the xy -plane moves along a straight line from A to B. How does the sign of $f_x(P)$ change? How does the sign of $f_y(P)$ change?

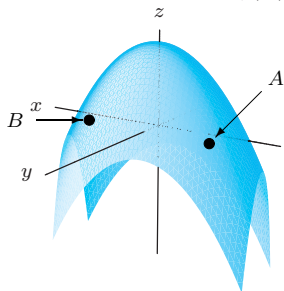


Figure 14.12

25. Figure 14.13 shows the saddle-shaped surface $z = f(x, y)$.

- (a) What is the sign of $f_x(0, 5)$?
 (b) What is the sign of $f_y(0, 5)$?

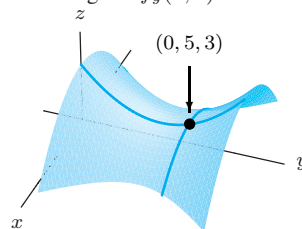


Figure 14.13

26. Figure 14.14 shows the graph of the function $f(x, y)$ on the domain $0 \leq x \leq 4$ and $0 \leq y \leq 4$. Use the graph to rank the following quantities in order from smallest to largest: $f_x(3, 2)$, $f_x(1, 2)$, $f_y(3, 2)$, $f_y(1, 2)$, 0.

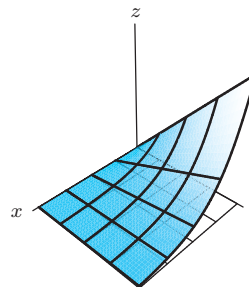


Figure 14.14

27. Figure 14.15 shows a contour diagram for the monthly payment P as a function of the interest rate, $r\%$, and the amount, L , of a 5-year loan. Estimate $\partial P / \partial r$ and $\partial P / \partial L$ at the following points. In each case, give the units and the everyday meaning of your answer.

- (a) $r = 8, L = 4000$ (b) $r = 8, L = 6000$
 (c) $r = 13, L = 7000$

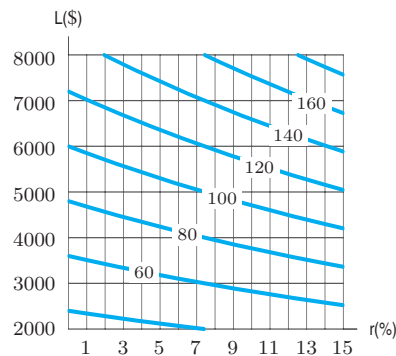


Figure 14.15

28. People commuting to a city can choose to go either by bus or by train. The number of people who choose either method depends in part upon the price of each. Let $f(P_1, P_2)$ be the number of people who take the bus when P_1 is the price of a bus ride and P_2 is the price of a train ride. What can you say about the signs of $\partial f / \partial P_1$ and $\partial f / \partial P_2$? Explain your answers.
29. When riding your bike in winter, the windchill temperature is a measure of how cold you feel as a result of the induced breeze caused by your travel. If W represents windchill temperature (in $^{\circ}\text{F}$) that you experience, then $W = f(T, v)$, where T is the actual air temperature (in $^{\circ}\text{F}$) and v is your speed, in meters per second. Match each of the practical interpretations below with a mathematical statement that most accurately describes it below. For the remaining mathematical statement, give a practical interpretation.

- (i) “The faster you ride, the colder you’ll feel.”
 (ii) “The warmer the day, the warmer you’ll feel.”

- (a) $f_T(T, v) > 0$
 (b) $f(0, v) \leq 0$
 (c) $f_v(T, v) < 0$

30. Figure 14.16 shows a contour diagram for the temperature T (in $^{\circ}\text{C}$) along a wall in a heated room as a function of distance x along the wall and time t in minutes. Estimate $\partial T / \partial x$ and $\partial T / \partial t$ at the given points. Give units and interpret your answers.

- (a) $x = 15, t = 20$ (b) $x = 5, t = 12$

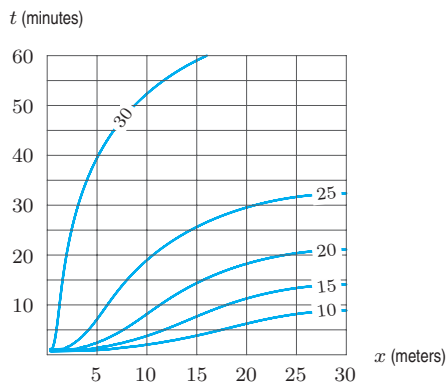


Figure 14.16

An airport can be cleared of fog by heating the air. The amount of heat required depends on the air temperature and the wetness of the fog. Problems 31–33 involve Figure 14.17, which shows the heat $H(T, w)$ required (in calories per cubic meter of fog) as a function of the temperature T (in degrees Celsius) and the water content w (in grams per cubic meter of fog). Note that Figure 14.17 is not a contour diagram, but shows cross-sections of H with w fixed at 0.1, 0.2, 0.3, and 0.4.

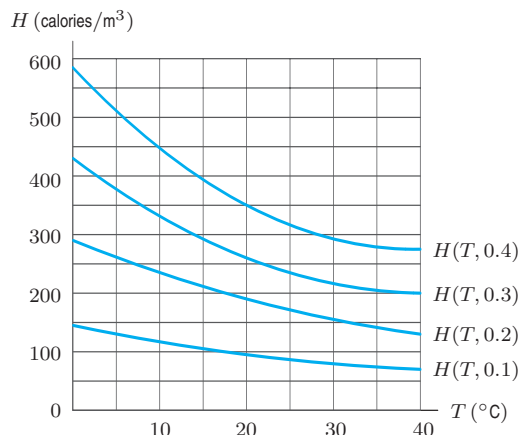


Figure 14.17

31. Use Figure 14.17 to estimate $H_T(10, 0.1)$. Interpret the partial derivative in practical terms.
32. Make a table of values for $H(T, w)$ from Figure 14.17, and use it to estimate $H_T(T, w)$ for $T = 10, 20$, and 30 and $w = 0.1, 0.2$, and 0.3 .
33. Repeat Problem 32 for $H_w(T, w)$ at $T = 10, 20$, and 30 and $w = 0.1, 0.2$, and 0.3 . What is the practical meaning of these partial derivatives?
34. The cardiac output, represented by c , is the volume of blood flowing through a person’s heart per unit time. The systemic vascular resistance (SVR), represented by s , is the resistance to blood flowing through veins and arteries. Let p be a person’s blood pressure. Then p is a function of c and s , so $p = f(c, s)$.

- (a) What does $\partial p / \partial c$ represent?

Suppose now that $p = kcs$, where k is a constant.

- (b) Sketch the level curves of p . What do they represent? Label your axes.
- (c) For a person with a weak heart, it is desirable to have the heart pumping against less resistance, while maintaining the same blood pressure. Such a person may be given the drug nitroglycerine to decrease the SVR and the drug Dopamine to increase the cardiac output. Represent this on a graph showing level curves. Put a point A on the graph representing the person’s state before drugs are given and a point B for after.
- (d) Right after a heart attack, a patient’s cardiac output drops, thereby causing the blood pressure to drop. A common mistake made by medical residents is to get the patient’s blood pressure back to normal by using drugs to increase the SVR, rather than by increasing the cardiac output. On a graph of the level curves of p , put a point D representing the patient before the heart attack, a point E representing the patient right

after the heart attack, and a third point F representing the patient after the resident has given the drugs to increase the SVR.

35. In each case, give a possible contour diagram for the

function $f(x, y)$ if

- (a) $f_x > 0$ and $f_y > 0$ (b) $f_x > 0$ and $f_y < 0$
 (c) $f_x < 0$ and $f_y > 0$ (d) $f_x < 0$ and $f_y < 0$

Strengthen Your Understanding

In Problems 36–37, explain what is wrong with the statement.

36. For $f(x, y)$, $\partial f / \partial x$ has the same units as $\partial f / \partial y$.
 37. The partial derivative with respect to y is not defined for functions such as $f(x, y) = x^2 + 5$ that have a formula that does not contain y explicitly.

In Problems 38–39, give an example of:

38. A table of values with three rows and three columns of a linear function $f(x, y)$ with $f_x < 0$ and $f_y > 0$.
 39. A function $f(x, y)$ with $f_x > 0$ and $f_y < 0$ everywhere.

Are the statements in Problems 40–49 true or false? Give reasons for your answer.

40. If $f(x, y)$ is a function of two variables and $f_x(10, 20)$ is defined, then $f_x(10, 20)$ is a scalar.
 41. If $f(x, y) = x^2 + y^2$, then $f_y(1, 1) < 0$.
 42. If the graph of $f(x, y)$ is a hemisphere centered at the origin, then $f_x(0, 0) = f_y(0, 0) = 0$.
 43. If $P = f(T, V)$ is a function expressing the pressure P (in grams/cm³) of gas in a piston in terms of the temperature T (in degrees °C) and volume V (in cm³), then $\partial P / \partial V$ has units of grams.
 44. If $f_x(a, b) > 0$, then the values of f decrease as we move in the negative x -direction near (a, b) .
 45. If $g(r, s) = r^2 + s$, then for fixed s , the partial derivative g_r increases as r increases.
 46. If $g(u, v) = (u + v)^u$, then $2.3 \leq g_u(1, 1) \leq 2.4$.
 47. Let $P = f(m, d)$ be the purchase price (in dollars) of a used car that has m miles on its engine and originally cost d dollars when new. Then $\partial P / \partial m$ and $\partial P / \partial d$ have the same sign.
 48. If $f(x, y)$ is a function with the property that $f_x(x, y)$ and $f_y(x, y)$ are both constant, then f is linear.
 49. If $f(x, y)$ has $f_x(a, b) = f_y(a, b) = 0$ at the point (a, b) , then f is constant everywhere.

14.2 COMPUTING PARTIAL DERIVATIVES ALGEBRAICALLY

Since the partial derivative $f_x(x, y)$ is the ordinary derivative of the function $f(x, y)$ with y held constant and $f_y(x, y)$ is the ordinary derivative of $f(x, y)$ with x held constant, we can use all the differentiation formulas from one-variable calculus to find partial derivatives.

Example 1 Let $f(x, y) = \frac{x^2}{y+1}$. Find $f_x(3, 2)$ algebraically.

Solution We use the fact that $f_x(3, 2)$ equals the derivative of $f(x, 2)$ at $x = 3$. Since

$$f(x, 2) = \frac{x^2}{2+1} = \frac{x^2}{3},$$

differentiating with respect to x , we have

$$f_x(x, 2) = \frac{\partial}{\partial x} \left(\frac{x^2}{3} \right) = \frac{2x}{3}, \quad \text{and so} \quad f_x(3, 2) = 2.$$

Example 2 Compute the partial derivatives with respect to x and with respect to y for the following functions.

- (a) $f(x, y) = y^2 e^{3x}$ (b) $z = (3xy + 2x)^5$ (c) $g(x, y) = e^{x+3y} \sin(xy)$

Solution (a) This is the product of a function of x (namely e^{3x}) and a function of y (namely y^2). When we differentiate with respect to x , we think of the function of y as a constant, and vice versa. Thus,

$$f_x(x, y) = y^2 \frac{\partial}{\partial x} (e^{3x}) = 3y^2 e^{3x},$$

$$f_y(x, y) = e^{3x} \frac{\partial}{\partial y} (y^2) = 2ye^{3x}.$$